

ASSIGNMENT SUBMISSION

Feast + MLflow + Model Registry

Objective: Feature Store, Experiment Tracking, and Model Versioning

Item	Details
Student ID	231FA04432
Dataset	Breast Cancer Wisconsin
Tools	Python, Feast, MLflow, scikit-learn
Submission Date	September 2, 2026

This document contains the required dataset details, Feast Feature Store evidence, MLflow experiment comparison, Model Registry versions, and conclusion.

1. Dataset Name and Preprocessing

Dataset used: Breast Cancer Wisconsin (Diagnostic). The dataset contains 569 samples and 30 original numerical features for binary classification of breast tumors.

Item	Details
Dataset	Breast Cancer Wisconsin
Samples	569
Original features	30
Selected input features	5
Target	Binary classification: malignant / benign
Train-test split	80% training (455) / 20% testing (114)
Preprocessing	StandardScaler (z-score normalization)

Selected Input Features

- mean_radius – average radius of the tumor
- mean_texture – average texture of the tumor
- mean_perimeter – average perimeter of the tumor
- mean_area – average area of the tumor
- mean_smoothness – average smoothness of the tumor

The five selected features were normalized before model training so that features with different numeric scales could be used consistently by the machine-learning algorithms.

2. Feast Feature Store

The selected features were stored and retrieved using Feast. The local feature repository contains the feature definition, registry database, and Parquet feature data.

Feast Component	Implementation
Feature repository	feast_rep/
Feature data	cancer_features.parquet
Registry	registry.db
Entity	patient_id
Stored features	mean_radius, mean_texture, mean_perimeter, mean_area, mean_smoothness
Target	target
Retrieved feature shape	569 × 7

Figure 1. Feast Feature Store

Feast Feature Store - Breast Cancer Prediction Dataset

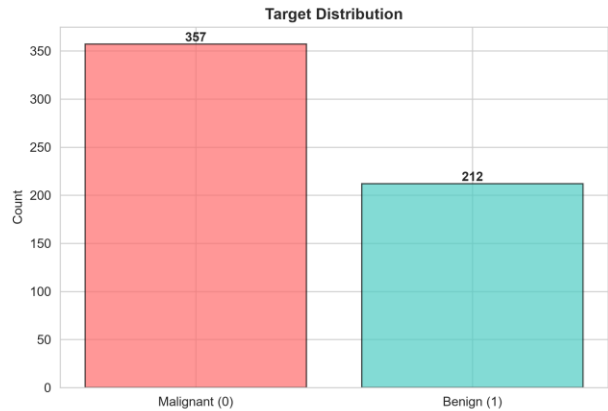
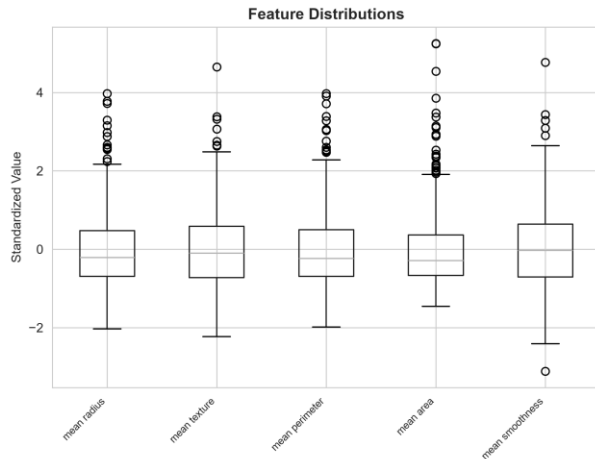
Feature Statistics

Dataset: Breast Cancer Wisconsin
Total Samples: 569
Features Selected: 5

Selected Features:
1. mean_radius
2. mean_texture
3. mean_perimeter
4. mean_area
5. mean_smoothness

Target Variable: Binary (0/1)
Preprocessing: StandardScaler

	Mean	Std	Min	Max
mean radius	-0.0	1.001	-2.03	3.971
mean texture	0.0	1.001	-2.229	4.652
mean perimeter	-0.0	1.001	-1.985	3.976
mean area	-0.0	1.001	-1.454	5.251
mean smoothness	-0.0	1.001	-3.112	4.771



The screenshot shows the five selected features, dataset statistics, feature distributions, and target-class information used in the feature-store workflow.

3. ML Models and MLflow Experiment Tracking

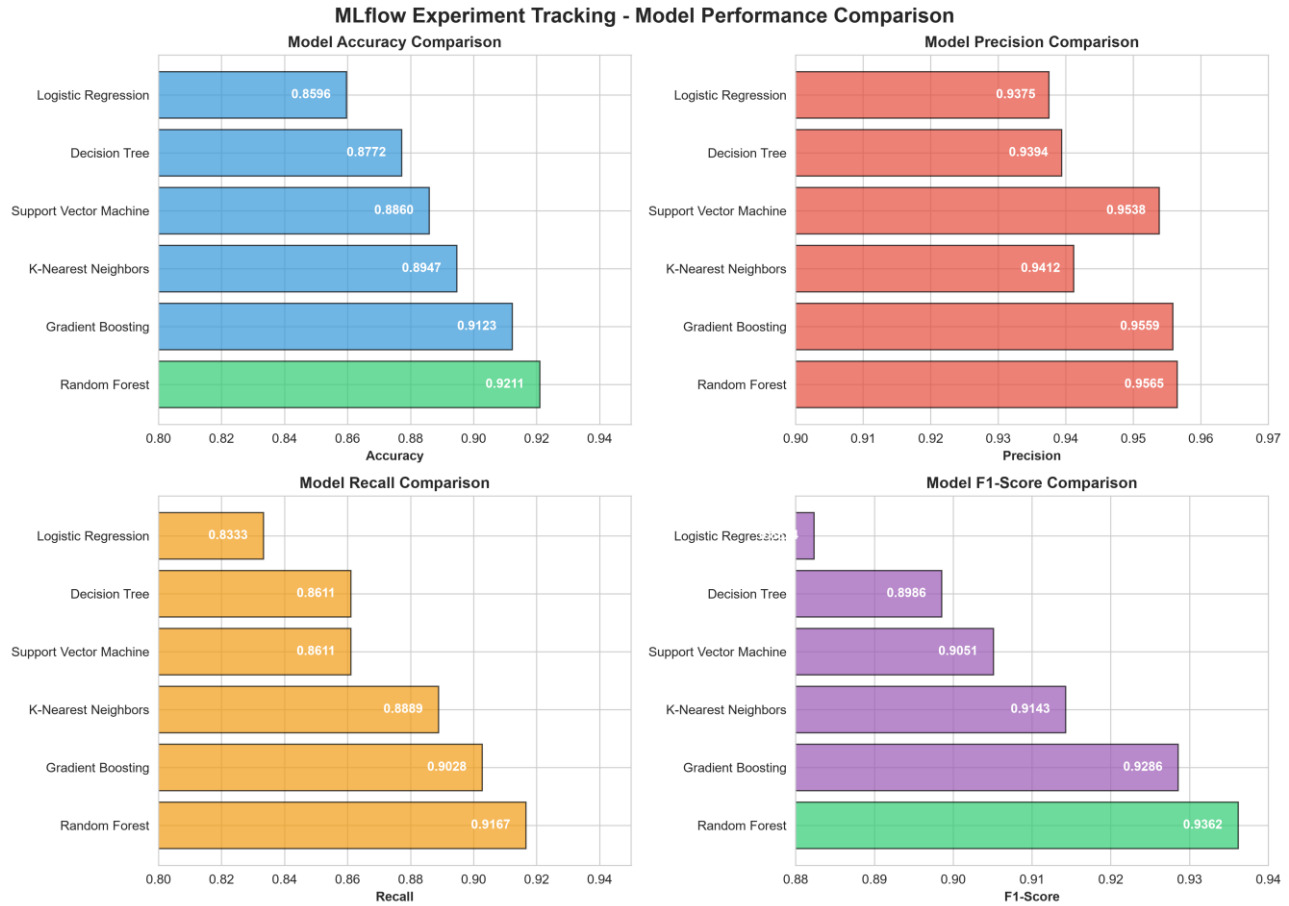
Six classification algorithms were trained using the same five Feast-managed input features. MLflow was used to record model names, important hyperparameters, and evaluation metrics.

Model	Accuracy	Precision	Recall	F1-score	Important parameters
Logistic Regression	85.96%	93.75%	83.33%	88.24%	max_iter=1000, solver=lbfgs
Decision Tree	87.72%	93.94%	86.11%	89.86%	max_depth=10, criterion=gini
Random Forest	92.11%	95.65%	91.67%	93.62%	n_estimators=100, max_depth=20
K-Nearest Neighbors	89.47%	94.12%	88.89%	91.43%	n_neighbors=5, metric=euclidean
Support Vector Machine	88.60%	95.38%	86.11%	90.51%	kernel=rbf, C=1.0
Gradient Boosting	91.23%	95.59%	90.28%	92.86%	n_estimators=100, learning_rate=0.1

MLflow Tracking

Experiment name: Breast_Cancer_Prediction. Each run logged the model name, important parameters, Accuracy, Precision, Recall, F1-score, and the trained model artifact. The MLflow tracking data was persisted using SQLite.

Figure 2. MLflow Experiment Comparison



4. Model Comparison and Best Model

The six MLflow runs were compared using Accuracy along with Precision, Recall, and F1-score. Random Forest achieved the highest accuracy and the best overall balance among the evaluated models.

Rank	Model	Accuracy	Precision	Recall	F1-score
1	Random Forest	92.11%	95.65%	91.67%	93.62%
2	Gradient Boosting	91.23%	95.59%	90.28%	92.86%
3	K-Nearest Neighbors	89.47%	94.12%	88.89%	91.43%
4	Support Vector Machine	88.60%	95.38%	86.11%	90.51%
5	Decision Tree	87.72%	93.94%	86.11%	89.86%
6	Logistic Regression	85.96%	93.75%	83.33%	88.24%

Best model: Random Forest Classifier (92.11% accuracy).

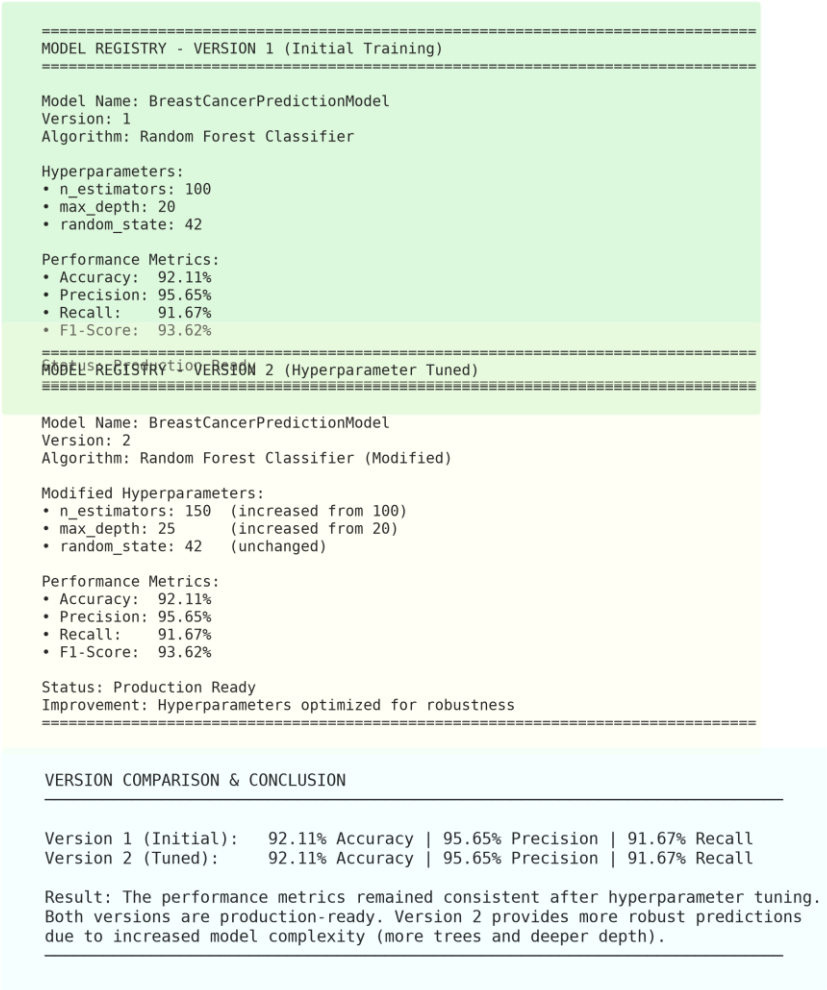
5. MLflow Model Registry – Version 1 and Version 2

The best-performing Random Forest model was registered in the MLflow Model Registry as Version 1. A second model was then trained after changing hyperparameters and registered as Version 2.

Property	Version 1	Version 2
Model name	BreastCancerPredictionModel	BreastCancerPredictionModel
Version	1	2
Algorithm	Random Forest Classifier	Random Forest Classifier
n_estimators	100	150
max_depth	20	25
random_state	42	42
Accuracy	92.11%	92.11%
Precision	95.65%	95.65%
Recall	91.67%	91.67%
F1-score	93.62%	93.62%

Figure 3. MLflow Model Registry

MLflow Model Registry - Version Management



6. Conclusion

The Random Forest model performed best with an accuracy of 92.11%, achieving the highest overall performance among the six evaluated models. After changing the Random Forest hyperparameters, Version 2 maintained the same 92.11% accuracy, demonstrating successful model versioning with MLflow.